Assignment 2

2024-03-25

R Markdown

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When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

```
#loading in the libraries
library(inspectdf)
```

```
## Warning: package 'inspectdf' was built under R version 4.3.3
```

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(forcats)
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.3.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v purrr   1.0.1     v tibble  3.2.1
## v readr   2.1.4     v tidyr   1.3.0
## v stringr 1.5.0
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
come errors
```

```
library(tidyr)
library(tinytex)
library(stringr)
library(readr)
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(moments)
```

```
ysn = 1895211
#calculating modulo
filenum <- (ysn + 1) %% 3
filenum
```

```
## [1] 1
```

```
filename <- paste0("//uofa//users$//users1//a1895211//Downloads//DT//atwo//gadget_1",filenu
m,".csv")
filename
```

```
## [1] "//uofa//users$//users1//a1895211//Downloads//DT//atwo//gadget_11.csv"
```

#Q1 Loading dataset and printing 10 lines

```
tibble_data <- read_csv("~//Downloads//DT//atwo/gadget_1.csv", show_col_types = FALSE)
head(n=10, tibble_data)
```

```
## # A tibble: 10 x 4
##    name         population advertising    sales
##    <chr>        <chr>      <chr>          <dbl>
##  1 Skillburgh   426637     $976,176.49    63072
##  2 Normouth     78983      $242,249.45    20378
##  3 Lunabrough   590032     $918,633.54    36460
##  4 Pineness     171642     $578,419.72    40302
##  5 Princefield  428436     $1,191,085.53  74462
##  6 Passland     459921     $1,353,578.27  98360
##  7 Chatpool     451084     $1,462,077.79 122668
##  8 Lexingcaster 295821     $907,909.89    71539
##  9 Springmouth  283199     $1,002,470.82  94711
## 10 Julville     386302     $816,050.82    58108
```

```
# adding row number column and shifting it to the left side
tibble_data <- tibble_data %>% mutate(row_number = row_number())%>% relocate(row_number)

head(tibble_data, n=10)
```

```
## # A tibble: 10 x 5
##    row_number name         population advertising    sales
##         <int> <chr>        <chr>      <chr>          <dbl>
##  1          1 Skillburgh   426637     $976,176.49    63072
##  2          2 Normouth     78983      $242,249.45    20378
##  3          3 Lunabrough   590032     $918,633.54    36460
##  4          4 Pineness     171642     $578,419.72    40302
##  5          5 Princefield  428436     $1,191,085.53  74462
##  6          6 Passland     459921     $1,353,578.27  98360
##  7          7 Chatpool     451084     $1,462,077.79 122668
##  8          8 Lexingcaster 295821     $907,909.89    71539
##  9          9 Springmouth  283199     $1,002,470.82  94711
## 10         10 Julville     386302     $816,050.82    58108
```

Q3 Identifying the variable types.

– row_number - It is just a label and has no ranking hence it is categorical nominal

– name - The name of each city is its unique identity so it is categorical nominal.

– population - It is quantitative discrete as population can be counted by an individual.

– advertising - It is measured in the accordance of money spent on advertising, so its Quantitative continuos.

–sales - is Quantitative Discrete, as sales can be counted.

Q 4 Cleaning and Taming data

```
#changing row_number to factor type

tibble_data$row_number <- as.factor(tibble_data$row_number)

# turning blank info in "NA" values

tibble_data[tibble_data == ""] <- NA

# checking the number of missing values
inspect_na(tibble_data)
```

```
## # A tibble: 5 x 3
##   col_name      cnt   pcnt
##   <chr>       <int>  <dbl>
## 1 name            3 0.300
## 2 population      1 0.0999
## 3 sales           1 0.0999
## 4 row_number      0 0
## 5 advertising     0 0
```

```
# Finding out the row number with missing name
which(is.na(tibble_data$name))
```

```
## [1]  45 237 870
```

```
#replacing "true" values with noname_x

tibble_data[45, ]$name <- "noname_1"
tibble_data[237, ]$name <- "noname_2"
tibble_data[870, ]$name <- "noname_3"

# Getting the duplicated names

n_col <- table(tibble_data$name)
n_num <- sort(n_col, decreasing = TRUE)
filter(tibble_data, name == "Kettleness")
```

```
## # A tibble: 2 x 5
##   row_number name       population advertising sales
##   <fct>      <chr>      <chr>      <chr>       <dbl>
## 1 49         Kettleness 160579     $264,199.70  5765
## 2 877        Kettleness 160579     $264,199.70  5765
```

```
filter(tibble_data, name == "Highside")
```

```
## # A tibble: 2 x 5
##   row_number name     population advertising sales
##   <fct>      <chr>    <chr>      <chr>       <dbl>
## 1 388        Highside 227818     $730,089.69 64004
## 2 594        Highside 227818     $730,089.69 64004
```

```
# removing the inital row with the same name

tibble_data <- filter(tibble_data, row_number !=50)
tibble_data <- filter(tibble_data, row_number != 389)

# name column to factor
tibble_data$name <- as.factor(tibble_data$name)
```

```
#cleaning and taming population column

pop <- sort(tibble_data$population, decreasing = TRUE)
index <- which(tibble_data$population == "two hundred and five thousand, three hundred and el
even")
tibble_data[259, ]$population <- "250311"

#Converting population type in integer to use abs function

tibble_data$population <- as.integer(tibble_data$population)
```

```
## Warning: NAs introduced by coercion
```

```
## Warning: NAs introduced by coercion to integer range
```

```
# Making population positive

tibble_data <- mutate(tibble_data, population = abs(population))
```

```
# Getting rid of useless symbols in advertising
tibble_data$advertising <- str_replace(tibble_data$advertising, ",", "")
tibble_data$advertising <- str_replace(tibble_data$advertising, "-", "")
tibble_data$advertising <- str_replace(tibble_data$advertising, "\\$", "")
tibble_data$advertising <- str_replace(tibble_data$advertising, ",", "")
```

```
# Filtering out the columns with NA values with filter function

tibble_data <- filter (tibble_data, !(is.na(population) | is.na(advertising) | is.na(sales)))
```

```
# changing the types of Sales and Advertising

tibble_data <- mutate(tibble_data, sales = abs(sales))
tibble_data$sales <- as.integer(tibble_data$sales)
```

```
## Warning: NAs introduced by coercion to integer range
```

```
tibble_data$advertising <- as.numeric(tibble_data$advertising)

# Filtering data and population, advertising and sales

tibble_data <- filter(tibble_data, population >= 10000, advertising >= 20000, sales >= 100)

# First 10 lines
head(tibble_data,10)
```

```
## # A tibble: 10 x 5
##    row_number name         population advertising  sales
##    <fct>      <fct>             <int>       <dbl>  <int>
##  1 1          Skillburgh       426637     976176.  63072
##  2 2          Normouth          78983     242249.  20378
##  3 3          Lunabrough       590032     918634.  36460
##  4 4          Pineness         171642     578420.  40302
##  5 5          Princefield      428436    1191086.  74462
##  6 6          Passland         459921    1353578.  98360
##  7 7          Chatpool         451084    1462078. 122668
##  8 8          Lexingcaster     295821     907910.  71539
##  9 9          Springmouth      283199    1002471.  94711
## 10 10         Julville         386302     816051.  58108
```

Q5 Create new variables in your tibble, representing the:

```
# percentage of a city's population who bought the gadget (call the variable sales pct)

tibble_data <- mutate(tibble_data, sales_pct = sales/population*100)

# advertising expenditure per person in a city (call the variable adv exp pp)

tibble_data <- mutate(tibble_data, adv_exp_pp = advertising/population)

head(tibble_data, 10)
```

```
## # A tibble: 10 x 7
##    row_number name         population advertising  sales sales_pct adv_exp_pp
##    <fct>      <fct>             <int>       <dbl>  <int>     <dbl>      <dbl>
##  1 1          Skillburgh       426637     976176.  63072     14.8        2.29
##  2 2          Normouth          78983     242249.  20378     25.8        3.07
##  3 3          Lunabrough       590032     918634.  36460      6.18       1.56
##  4 4          Pineness         171642     578420.  40302     23.5        3.37
##  5 5          Princefield      428436    1191086.  74462     17.4        2.78
##  6 6          Passland         459921    1353578.  98360     21.4        2.94
##  7 7          Chatpool         451084    1462078. 122668     27.2        3.24
##  8 8          Lexingcaster     295821     907910.  71539     24.2        3.07
##  9 9          Springmouth      283199    1002471.  94711     33.4        3.54
## 10 10         Julville         386302     816051.  58108     15.0        2.11
```

– Sales_pct -> is a percentage which means it can be measured hence it's Quantitative Continuous – adv_exp_pp -> can also be measured hence it is quantitative continuous

Q6 Set seed and take a random sampe of 700 cities

```
set.seed(ysn)
r_sample <- sample_n(tibble_data, 700)
head(r_sample, 10)
```

```
## # A tibble: 10 x 7
##    row_number name        population advertising  sales sales_pct adv_exp_pp
##    <fct>      <fct>            <int>       <dbl>  <int>     <dbl>      <dbl>
##  1 876        Ashwich         156949     410057.  25664     16.4        2.61
##  2 139        Factcaster      340725    1235172. 107253     31.5        3.63
##  3 241        Costsland       125699     542105.  60730     48.3        4.31
##  4 409        Chatport        642867    1669547. 114493     17.8        2.60
##  5 453        Fauxport        205588     415570.  24359     11.8        2.02
##  6 285        Bellgrad        506427    1044595.  36554      7.22       2.06
##  7 255        Kettletown      425174     952510.  45436     10.7        2.24
##  8 236        Passcaster      267048     925342.  86092     32.2        3.47
##  9 327        Sugarbrough     235512     547838.  36739     15.6        2.33
## 10 943        Passstead       352962    1347602. 108497     30.7        3.82
```

#Q7 Use inspect num

```
options(scipen = 999, digits = 7)
inspect_num(r_sample)
```

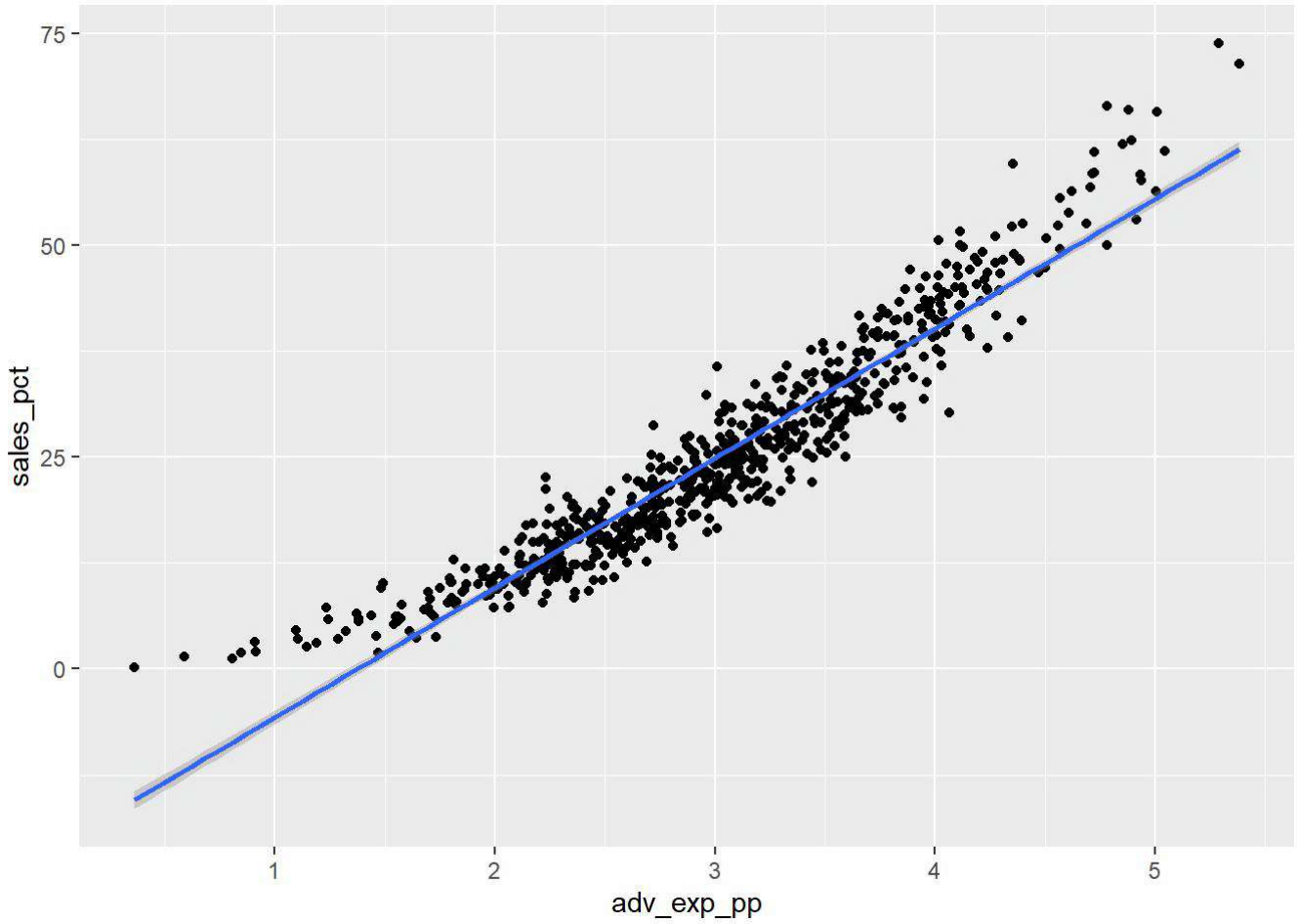
```
## # A tibble: 5 x 10
##   col_name       min     q1 median   mean     q3    max      sd pcnt_na hist
##   <chr>        <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>   <dbl>   <dbl> <named >
## 1 population 2.18e+4 2.16e5 3.28e5 3.59e5 4.79e5 1.22e6 1.94e+5       0 <tibble>
## 2 advertisi~ 5.52e+4 5.87e5 9.59e5 1.09e6 1.45e6 3.64e6 6.71e+5       0 <tibble>
## 3 sales      7.77e+2 3.98e4 7.18e4 9.06e4 1.19e5 4.53e5 7.02e+4       0 <tibble>
## 4 sales_pct  1.45e-1 1.57e1 2.35e1 2.52e1 3.26e1 7.38e1 1.29e+1       0 <tibble>
## 5 adv_exp_pp 3.61e-1 2.47e0 3.02e0 3.02e0 3.56e0 5.38e0 7.99e-1       0 <tibble>
```

Q8 Produce a scatterplot of the two new columns sales pct and adv exp pp, with the explanatory variable on the horizontal axis and the response variable on the vertical axis. Include a straight line of best fit on your

plot. Does it look like there is a linear relationship between the two variables?

```
ggplot(r_sample, aes(x = adv_exp_pp, y = sales_pct)) +
  geom_point()+
  geom_smooth(method = lm)
```

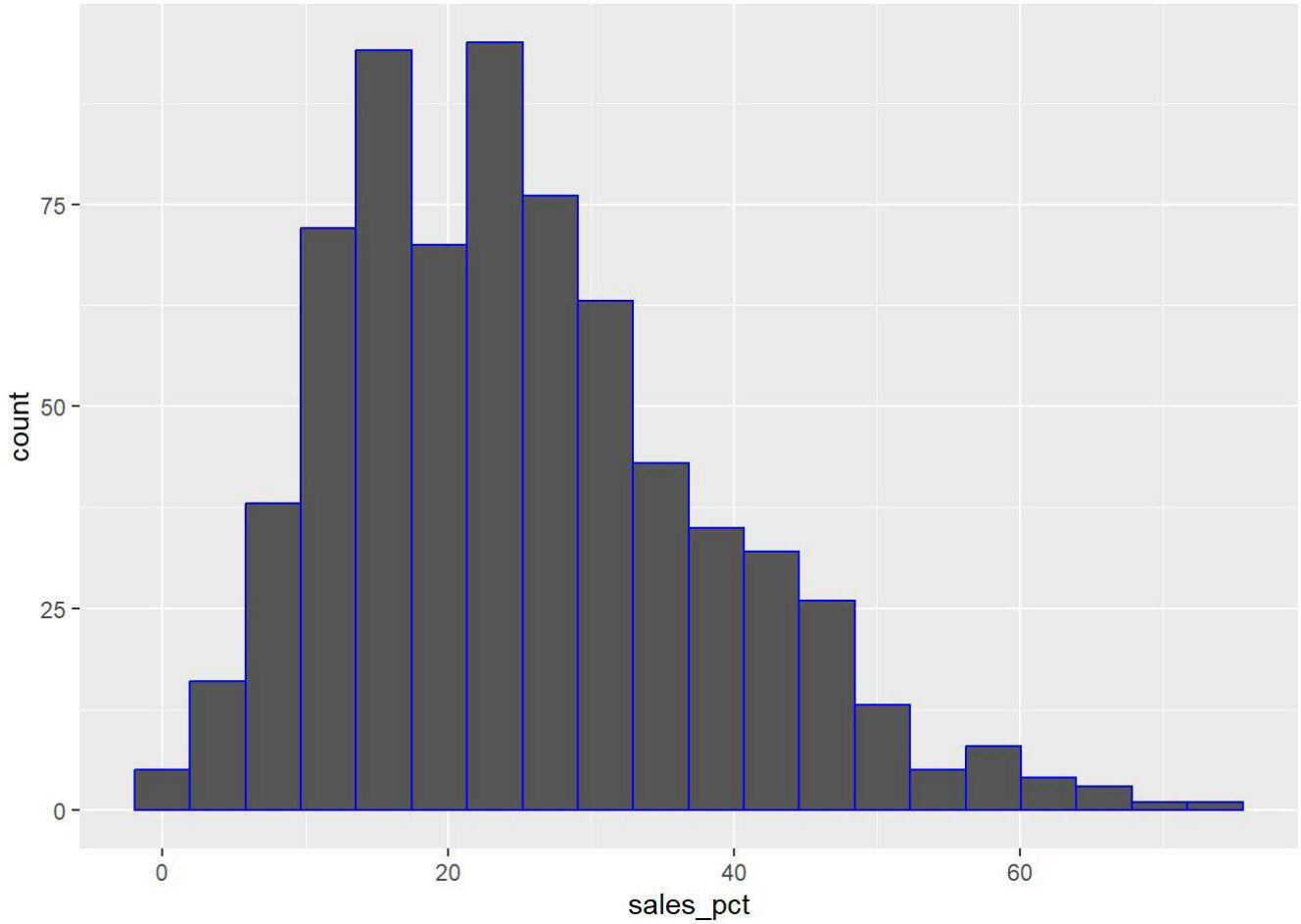
```
## `geom_smooth()` using formula = 'y ~ x'
```



There is a curve in the graph which indicates that there is no linear relationship in the scatter plot

#Q9 Plot a histogram of sales pct and calculate the skewness. Does the data look like a standard normal distribution?

```
ggplot(r_sample, aes(sales_pct))+
  geom_histogram(bins = 20, col = "blue")
```




```
skewness(r_sample$sales_pct)
```

```
## [1] 0.7168786
```

– The distribution is not a standard normal distribution as skewness value is 0.7168786 which means the distribution is right-skewed and can be seen in the graph aswell.

Q 10

```
# BoxCoxTrans to obtain an estimate of λ. What is the estimate?
bct <- BoxCoxTrans(y = r_sample$sales_pct)
bct
```

```
## Box-Cox Transformation
##
## 700 data points used to estimate Lambda
##
## Input data summary:
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.1455 15.7304 23.4531 25.1806 32.6163 73.8130
##
## Largest/Smallest: 507
## Sample Skewness: 0.715
##
## Estimated Lambda: 0.5
```

The estimate value of λ is 0.5

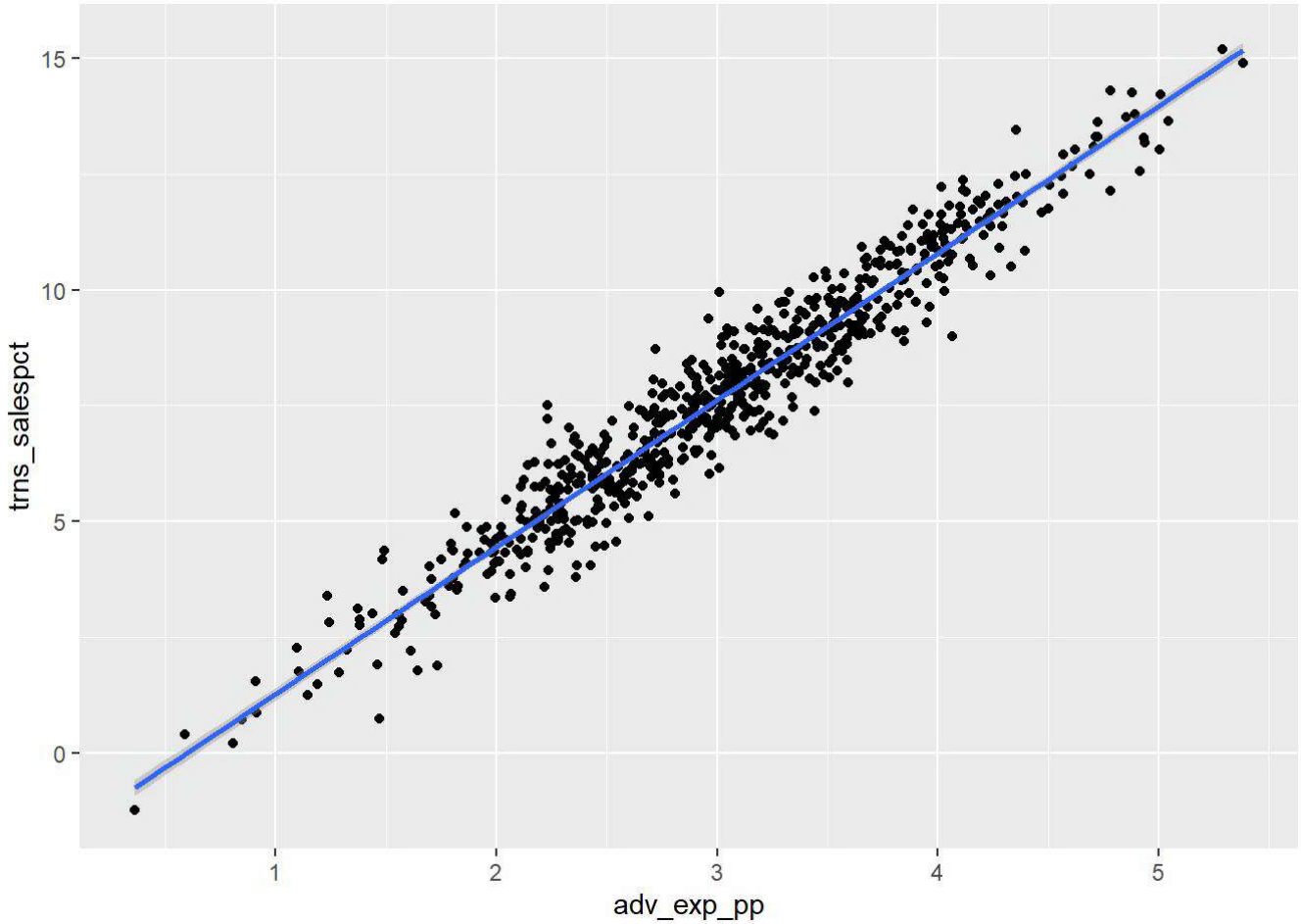
```
# Apply transformation to create new column
r_sample <- mutate(
  r_sample, trns_salespct = predict(bct, sales_pct))
head(r_sample, 10)
```

```
## # A tibble: 10 x 8
##    row_number name        population advertising  sales sales_pct adv_exp_pp
##    <fct>      <fct>            <int>       <dbl>  <int>     <dbl>      <dbl>
##  1 876        Ashwich         156949     410057.  25664     16.4        2.61
##  2 139        Factcaster      340725    1235172. 107253     31.5        3.63
##  3 241        Costsland       125699     542105.  60730     48.3        4.31
##  4 409        Chatport        642867    1669547. 114493     17.8        2.60
##  5 453        Fauxport        205588     415570.  24359     11.8        2.02
##  6 285        Bellgrad        506427    1044595.  36554      7.22       2.06
##  7 255        Kettletown      425174     952510.  45436     10.7        2.24
##  8 236        Passcaster      267048     925342.  86092     32.2        3.47
##  9 327        Sugarbrough     235512     547838.  36739     15.6        2.33
## 10 943        Passstead       352962    1347602. 108497     30.7        3.82
## # i 1 more variable: trns_salespct <dbl>
```

Q11

```
# scatterplot
ggplot(r_sample, aes(x = adv_exp_pp, y = trns_salespct)) +
  geom_point()+
  geom_smooth(method = "lm")
```

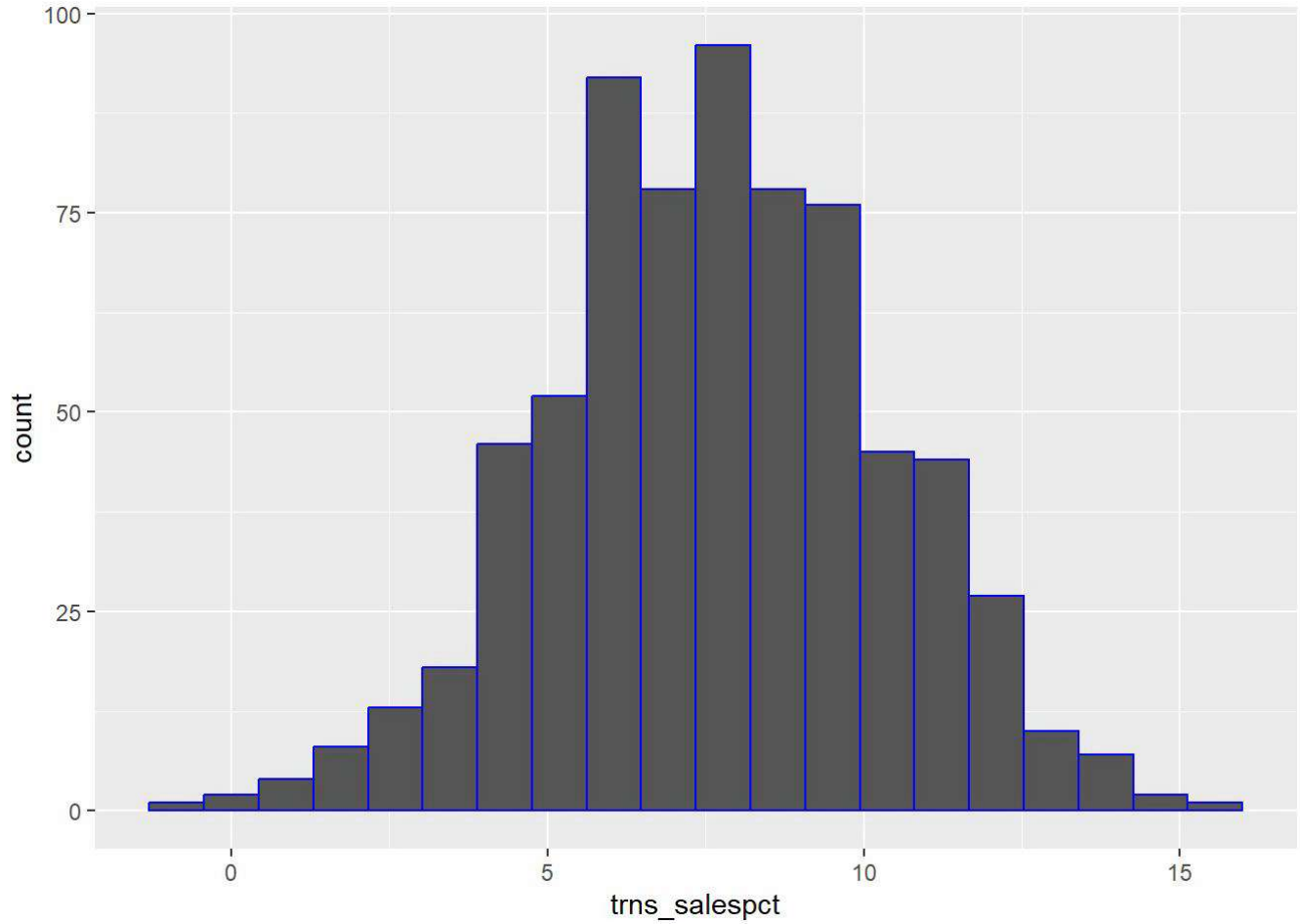
```
## `geom_smooth()` using formula = 'y ~ x'
```



This looks like a strong,positive linear scatterplot and it has some outliers at (7.5, 2.4) and (10, 3)

histogram

```
ggplot(r_sample, aes(trns_salespct))+
  geom_histogram(bins = 20, col = "blue")
```




```
skewness(r_sample$trns_salespct)
```

```
## [1] -0.04523431
```

The skewness value of approximately -0.04523431 suggests a slight left-skewness in the distribution.

Q 12_1

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Q 12_2

y_i = response variable(trns_salespct) β_0 = intercept(adv_exp_pp) β_1 = slope ϵ_i= error term

$$y_i = \beta_0 + \beta_1 x_i$$

y_i = trns_salespct estimate x_i = adv_exp_pp observed value $beta_0$ = intercept $beta_1$ = slope

Q12_3rd

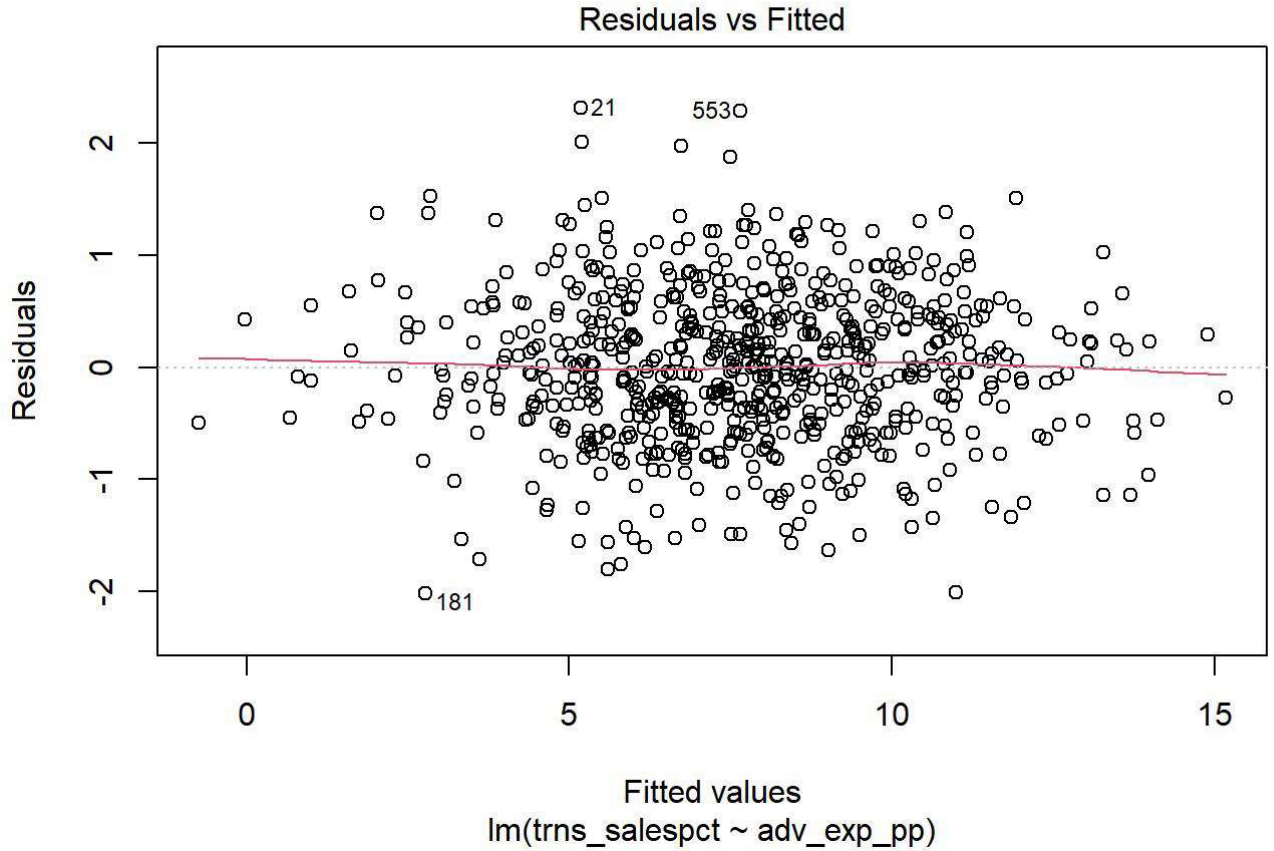
```
tibble_slope <- lm(trns_salespct ~ adv_exp_pp, data = r_sample)
summary(tibble_slope)
```

```
##
## Call:
## lm(formula = trns_salespct ~ adv_exp_pp, data = r_sample)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.02268 -0.44927  0.00334  0.45204  2.31538
##
## Coefficients:
##             Estimate Std. Error t value            Pr(>|t|)
## (Intercept) -1.88945    0.10085  -18.74 <0.0000000000000002 ***
## adv_exp_pp   3.17091    0.03228   98.23 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.682 on 698 degrees of freedom
## Multiple R-squared:  0.9325, Adjusted R-squared:  0.9324
## F-statistic:  9649 on 1 and 698 DF,  p-value: < 0.00000000000000022
```

Y = -1.88 + 3.17 Y = trns_salespct x = adv_exp_pp

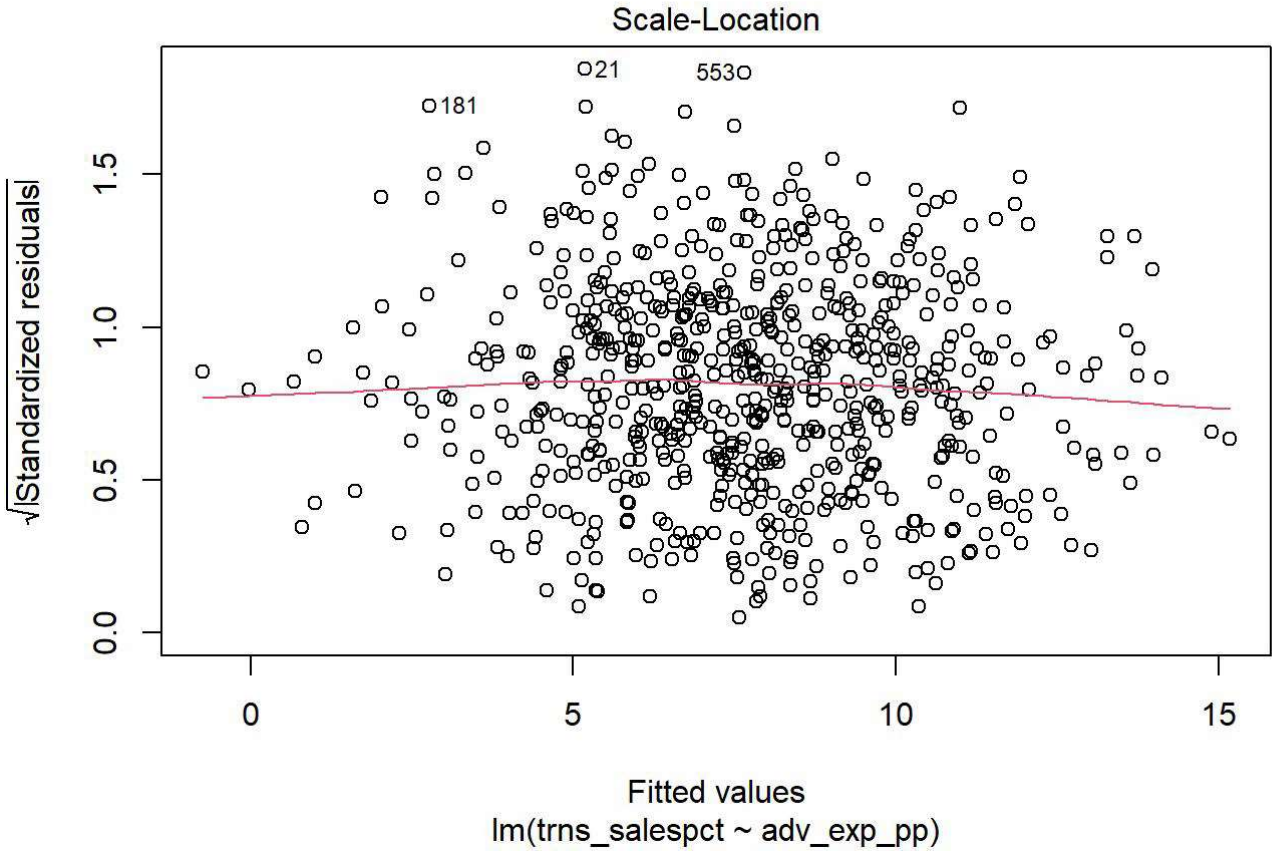
Q13 Using the material in Module 6 (p. 10–13), check if our model satisfies the 4 assumptions for a linear model.You will need to write some text explaning your conclusions here. Make sure you identify at least one possible problem with the Independence assumption.

```
plot(tibble_slope, which = 1)
```



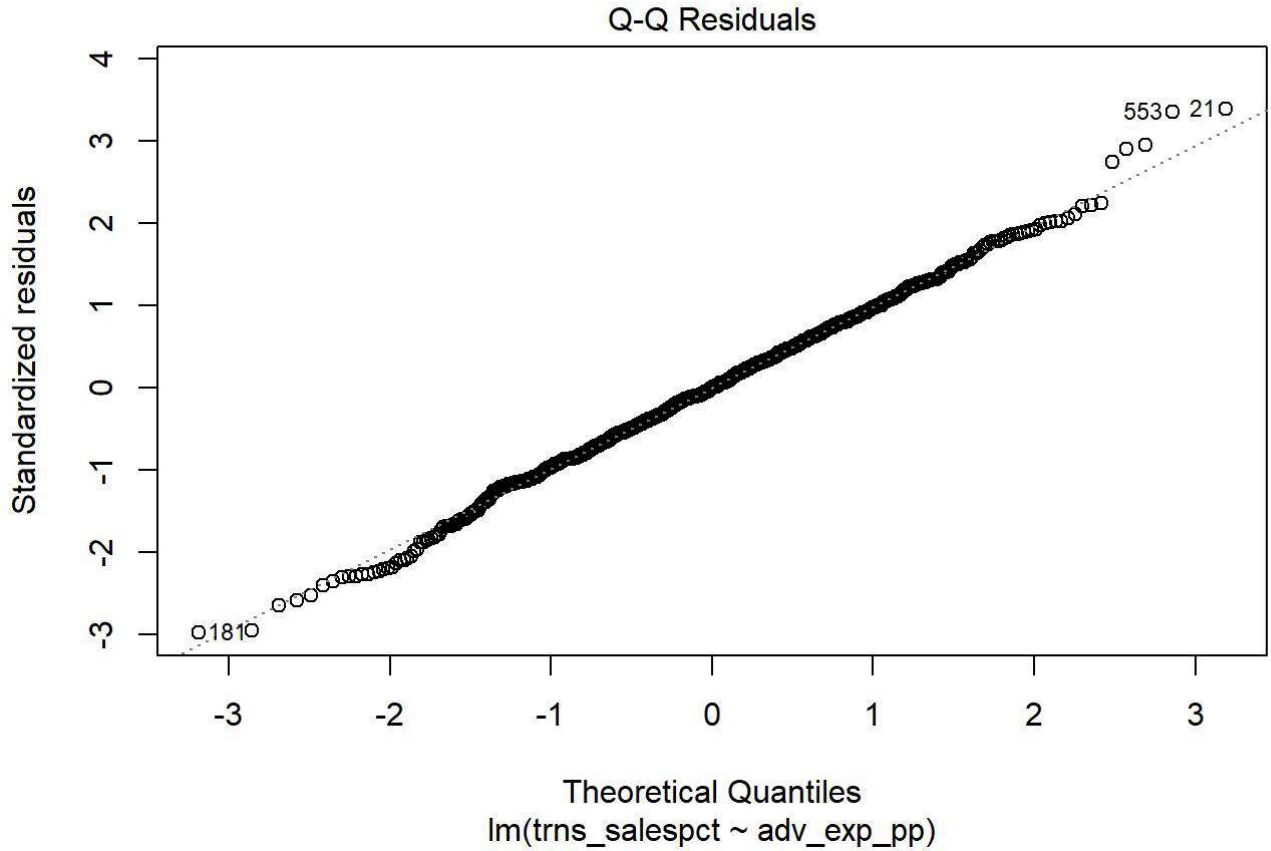
There cant be seen any specific trends in the plot so the assumption can be said as good.

```
plot(tibble_slope, which = 3)
```



There are no apparent trends here .The model has constant spread. we can assume the noise structure here is also good.

```
#normality
plot(tibble_slope, which = 2)
```



Most of the points are close to dotted lines and they are distributed mostly normal.

Independance

the assumption of independent refer to data points that are independent of each other. It is not possible here because the population might keep changing with the cities they move to and buy stuff over there which affects the sales of the cities too.

#Q14

```
lambda <- bct$lambda

ls = tibble(adv_exp_pp = 0.05)
ls_predict <- predict(tibble_slope, ls, interval = "prediction", level = 0.7)
ls_predict_r <- (lambda*ls_predict+1)^(1/lambda)
ls_predict_r
```

```
##          fit        lwr       upr
## 1 0.01810306 0.04966319 0.2420124
```

0.018% of city population will buy Gadget 2 with the interval 0.049% to 0.242%

Medium spend

```
ms <- tibble(adv_exp_pp = 3.14)
mp <- predict(tibble_slope, ms, interval = "prediction", level = 0.9)
mpr <- (lambda*mp+1)^(1/lambda)
mpr
```

```
##       fit      lwr    upr
## 1 25.3371 19.99495 31.311
```

25.33 % city population will buy Gadget two, Interval from 20% to 31.31% # high spend

```
hs <- tibble(adv_exp_pp = 6)
hp <- predict(tibble_slope, hs, interval = "prediction", level = 0.9)
hpr <- (lambda*hp+1)^(1/lambda)
hpr
```

```
##        fit      lwr      upr
## 1 91.54649 81.00757 102.7297
```

91. 53% of city population will buy Gadget 2 with the interval from 81% to 102.72%

Q15

As we can see the above predictions, It states that the product will have more sales if alot of advertisement is done. According to the results of the 3 level of money spends on Product, if the company spends $6 dollars per person for the advertisement it yeilds in a a staggering 91.54% of population buying Gadget 2.

On the other hand there is not a proper outcome in the low spend prediction. Reason behind this could be some outliers.